The Journey of Building
"A Trip On The Bloch Sphere"

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1 Introduction

This document chronicles my journey in creating **A Trip on The Bloch Sphere**. It captures my thought process, all my fails, successes, and ideas for the future to come.

2 The Math

Since finding the probability amplitudes of a qubit without explicit calculation is impossible, it is necessary to use matrix multiplication to find the state after each operation applied. While I could use Python or a library that already implements matrix multiplication, what's the fun in that?
So the first step is to make a library that implements matrix multiplication, and what better language to do it in than C? It's fast, simple, and straightforward. It has all I need: no more, no less.
The first step is to use a struct to implement complex numbers. It simply has to have a real part and imaginary part.
Now for the

2.1 Converting Vector to Spherical Coordinates

We can represent any qubit on the Bloch Sphere using the spherical coordinates (θ, ϕ, r). Since $|\alpha^2| + |\beta^2| = 1$, r is always 1 and so for any pure state qubit, the resulting coordinate will always be on the surface of the Bloch Sphere.
Any qubit can be represented in the following form: $\cos(\frac{\theta}{2})|0\rangle + e^{i\phi}\sin(\frac{\theta}{2})|1\rangle$, where ϕ is the relative phase between $|0\rangle$ and $|1\rangle$. θ will be limited to values between 0 and π, so the fact that cos is a symetric function does not matter: the negative value of θ can always be rejected.
Now that we know the value of θ, we can plug it into $e^{i\phi}\sin(\frac{\theta}{2}) = \beta$ to solve for the only unkown variable: ϕ.
Solving this gives us: $\phi = -i \cdot \ln(\frac{\beta}{\sin(\frac{\theta}{2})})$, where we use $-i$ as the inverse of i, and use $\phi = 0$ if $\theta = 0$.

Now going back to the part about ϕ being the relative phase, we can easily make sure this assumption always holds true by factoring out any complex number